



COMSATS University Islamabad,
Sahiwal Campus

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

ReviewPilot

Version 1.0

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Revision History

Name	Date	Reason for changes	Version
Manal Hussain	30-Sep-2026	Initial SRS document	1.0
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Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

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1. Introduction

This Software Requirements Specification (SRS) describes the requirements of ReviewPilot, an AI-powered web application that helps local businesses respond to their Google reviews. It is organized as follows: Section 2 gives an overall description of the product; Section 3 explains the requirement identifying technique and presents the use cases; Section 4 lists the specific functional requirements by feature; Section 5 states the quality attributes; Section 6 describes the external interfaces; Section 7 contains the project Gantt chart; and Section 8 lists the references. The intended readers are the development team, the project supervisor, and the evaluation committee.

1.1 Purpose

The purpose of this document is to identify and specify the requirements of the product ReviewPilot, version 1.0, so that the design, implementation, and testing of the system can be carried out against a clear and agreed baseline.

1.2 Scope

ReviewPilot is a full-stack web application built with the MERN stack (MongoDB, Express.js, React, Node.js) and the Google Gemini API. Its purpose is to reduce the effort a business owner spends on answering Google reviews while keeping the owner in control of what is published. The major features are: user authentication; linking a Google Business Profile with OAuth 2.0; managing business locations and response settings; automatic fetching of new reviews every 30 minutes; sentiment classification and AI-drafted replies; owner approval, editing, regeneration, or skipping of replies; posting approved replies to Google; an analytics dashboard; email notifications; and basic administration. The first version does not support platforms other than Google, payment processing or billing, non-English replies, or a native mobile application.

2. Overall description

2.1 Product perspective

ReviewPilot is an entirely new, standalone web-based (SaaS) product. It is not a replacement for or an extension of an existing application. It acts as an intermediary between the business owner and the owner's Google reviews and adds an AI layer that drafts replies. It interfaces with the following external systems:

- Google Business Profile API: to fetch reviews and post replies.
- Google OAuth 2.0: to authorize access to the owner's profile.
- Google Gemini API: to classify sentiment and generate replies.
- MongoDB Atlas: for persistent data storage.
- SMTP email service (Nodemailer): to send notifications.

2.2 Operating environment

OE-1: The system shall operate correctly on the following web browsers: Google Chrome 90+, Mozilla Firefox 90+, Microsoft Edge 90+, and Apple Safari 15+.

OE-2: The server-side application shall run on Node.js version 18 or later.

OE-3: The database shall be MongoDB 6.0 or later, hosted on MongoDB Atlas.

OE-4: The frontend shall be hosted on a cloud platform such as Vercel and the backend on a platform such as Render or Railway.

OE-5: All communication shall use HTTPS, and the user interface shall work on screen widths from 375 px (mobile) to 1920 px (desktop).

2.3 Design and implementation constraints

CO-1: The system shall use the MERN stack (MongoDB, Express.js, React, Node.js), as required by the course.

CO-2: The system shall handle Google Business Profile API daily quota limits gracefully.

CO-3: The system shall keep prompts short and minimize Gemini calls because API usage is limited by quota and cost.

CO-4: Use of the Google Business Profile API requires business verification and approval, which may cause delays.

CO-5: The system shall comply with Google's OAuth 2.0 policies, including the consent screen requirements.

CO-6: User credentials and Google tokens shall be stored securely (encrypted at rest).

CO-7: Because the project has no budget, only free tiers of services shall be used.

3. Requirement identifying technique

ReviewPilot is an interactive, end-user application, so requirements are identified using the use case technique. Use cases capture the goals of each actor and are used to derive the functional requirements in Section 4. The actors are the Business Owner, the System Administrator, the Scheduler (cron job), and the external systems Google Business Profile API and Google Gemini API.

3.1 Use Case Diagram

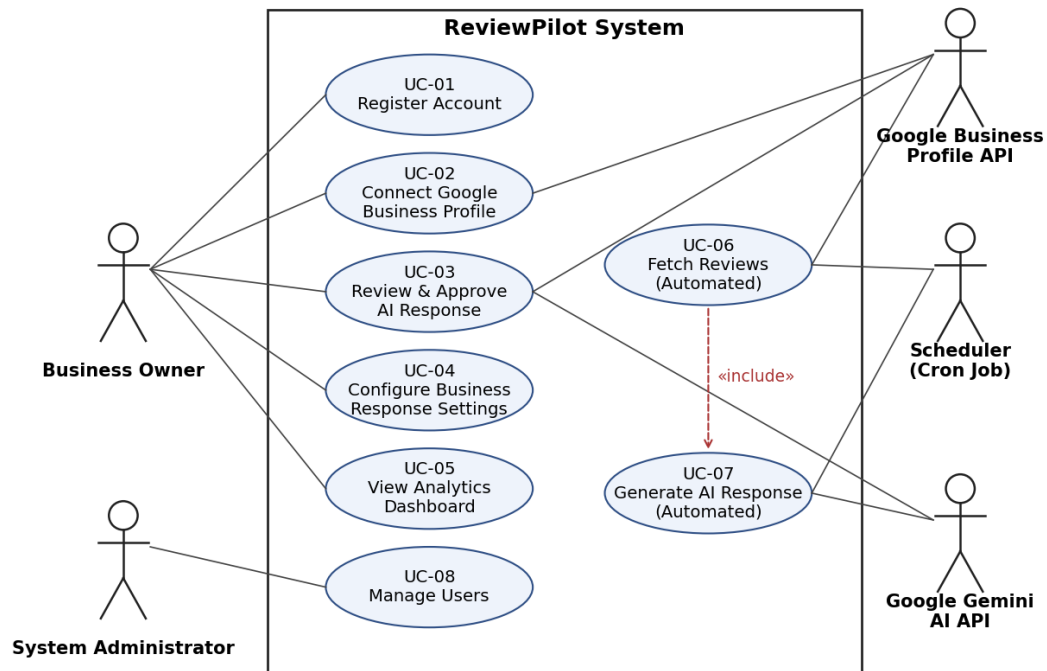


Figure 1 Use case diagram of ReviewPilot

3.2 Use Case Description

The following tables describe each use case using the template of the SRS guideline.

Table 5.1 Use case description: UC-01 Register Account

Use Case ID	UC-01
Use Case Name	Register Account
Actors	Primary Actor: Business Owner Secondary Actors: None
Description	A new user creates a ReviewPilot account by providing a name, email, and password.
Trigger	A visitor selects Register.
Preconditions	PRE-1. User has a valid email address. PRE-2. User is not already registered.
Postconditions	POST-1. User account is created with the role business_owner. POST-2. User is logged in with a JWT and redirected to the dashboard.
Normal Flow	1.0 Register Account 1. User opens the registration page. 2. User enters name, email, and password. (see 1.0.E1) 3. System validates the input. 4. System checks that the email is not registered. (see 1.0.E2) 5. System hashes the password and creates the user record.

	6. System generates a JWT. 7. System returns the token and redirects the user to the dashboard.
Alternative Flows	None.
Exceptions	1.0.E1 Invalid input 1. System displays validation errors. 2. User corrects the input and returns to step 2. 1.0.E2 Email already registered 1. System displays "Email already registered". 2. User enters a different email or goes to login.
Business Rules	BR-1 Email addresses must be unique. BR-2 Passwords must be at least 6 characters.
Assumptions	1. Users have a modern browser with JavaScript enabled.

Table 5.2 Use case description: UC-02 Connect Google Business Profile

Use Case ID	UC-02
Use Case Name	Connect Google Business Profile
Actors	Primary Actor: Business Owner Secondary Actors: Google OAuth Service, Google Business Profile API
Description	The owner links a Google Business Profile so ReviewPilot can read reviews and post replies, then selects which locations to monitor.
Trigger	The owner selects Connect Google Business Profile.
Preconditions	PRE-1. Owner is logged in. PRE-2. Owner has a Google Business Profile.
Postconditions	POST-1. Google tokens are stored encrypted. POST-2. Selected business locations are created and review fetching begins.
Normal Flow	2.0 Connect Google Business Profile 1. Owner selects Connect Google Business Profile. 2. System generates the Google OAuth URL with the required scopes. 3. System redirects the owner to Google's consent screen. 4. Owner grants permission. (see 2.0.E1) 5. Google redirects to the ReviewPilot callback with an authorization code. 6. System exchanges the code for access and refresh tokens. (see 2.0.E2) 7. System stores the tokens encrypted. 8. System fetches the available business locations. 9. Owner selects the locations to monitor. 10. System creates the business records and starts fetching reviews.
Alternative Flows	None.
Exceptions	2.0.E1 Owner denies permission 1. System displays "Connection cancelled" and terminates the use case. 2.0.E2 Token exchange fails

	1. System displays an error and asks the owner to try again.
Business Rules	BR-1 Access must comply with Google OAuth 2.0 policies. BR-2 Tokens must be stored encrypted.
Assumptions	1. The Google Business Profile API remains available to the project.

Table 5.3 Use case description: UC-03 Review and Approve AI Response

Use Case ID	UC-03
Use Case Name	Review and Approve AI Response
Actors	Primary Actor: Business Owner Secondary Actors: Google Business Profile API
Description	The owner reads a new review and its AI-drafted reply and approves, edits, regenerates, or skips it. Approved replies are posted to Google.
Trigger	The owner opens a pending review on the dashboard.
Preconditions	PRE-1. Owner is logged in. PRE-2. A new review has been fetched and an AI reply has been generated.
Postconditions	POST-1. The review status is posted (or skipped). POST-2. The approved reply is visible on Google.
Normal Flow	3.0 Review and Approve AI Response 1. Owner opens the dashboard and sees pending reviews. 2. Owner reads the customer's review. 3. Owner reads the AI-generated reply. (see 3.1, 3.2, 3.3) 4. Owner selects Approve. 5. System posts the reply to Google. (see 3.0.E1) 6. System updates the review status to posted. 7. System moves the review from pending to responded.
Alternative Flows	3.1 Edit reply 1. Owner selects Edit and changes the text. 2. Owner selects Save and Approve; return to step 5 of normal flow. 3.2 Regenerate reply 1. Owner selects Regenerate. 2. System generates a new reply; return to step 3 of normal flow. 3.3 Skip review 1. Owner selects Skip. 2. System marks the review as skipped and terminates the use case.
Exceptions	3.0.E1 Posting to Google fails 1. System informs the owner and keeps the status as approved. 2. Owner may retry posting later.
Business Rules	BR-1 Only approved or edited replies may be posted. BR-2 A review may be posted only once.
Assumptions	1. Owners check the dashboard or email notifications regularly.

Table 5.4 Use case description: UC-04 Configure Business Response Settings

Use Case ID	UC-04
Use Case Name	Configure Business Response Settings
Actors	Primary Actor: Business Owner Secondary Actors: None
Description	The owner customizes how AI replies are generated for a business.
Trigger	The owner opens Business Settings.
Preconditions	PRE-1. Owner is logged in. PRE-2. The business is connected and active.
Postconditions	POST-1. Updated settings are saved and applied to future AI replies.
Normal Flow	4.0 Configure Business Response Settings 1. Owner opens Business Settings. 2. Owner selects the response tone. 3. Owner enters the sign-off name. 4. Owner adds SEO keywords. 5. Owner selects the response language. 6. Owner optionally enables auto-approve for 4-5 star reviews. 7. Owner selects Save Settings. (see 4.0.E1) 8. System validates and saves the configuration.
Alternative Flows	None.
Exceptions	4.0.E1 Invalid input 1. System displays validation errors and returns to step 2.
Business Rules	BR-1 Tone must be professional, friendly, casual, or formal. BR-2 Language is English only in this version.
Assumptions	1. Owners know their preferred tone and keywords.

Table 5.5 Use case description: UC-05 View Analytics Dashboard

Use Case ID	UC-05
Use Case Name	View Analytics Dashboard
Actors	Primary Actor: Business Owner Secondary Actors: None
Description	The owner views review performance metrics and charts.
Trigger	The owner opens the Analytics page.
Preconditions	PRE-1. Owner is logged in. PRE-2. At least one business is connected with fetched reviews.
Postconditions	POST-1. None (read-only).

Normal Flow	5.0 View Analytics Dashboard 1. Owner opens the Analytics page. 2. System retrieves aggregated data. (see 5.0.E1) 3. System displays total reviews, pending responses, response rate, and average rating. 4. System displays the review volume line chart, sentiment donut chart, and rating distribution bar chart. 5. Owner filters by business or changes the date range (see 5.1).
Alternative Flows	5.1 Change filter 1. Owner selects a business or date range. 2. Return to step 2 of normal flow.
Exceptions	5.0.E1 No reviews yet 1. System displays an empty state message.
Business Rules	BR-1 $\text{Response rate} = \frac{\text{responded reviews}}{\text{total reviews}} \times 100$.
Assumptions	1. Data is at most one fetch cycle (30 minutes) old.

Table 5.6 Use case description: UC-06 Fetch Reviews (Automated)

Use Case ID	UC-06
Use Case Name	Fetch Reviews (Automated)
Actors	Primary Actor: Scheduler (Cron Job) Secondary Actors: Google Business Profile API
Description	The system automatically retrieves new Google reviews for every active business and stores them.
Trigger	The 30-minute schedule fires (or the owner requests a manual fetch).
Preconditions	PRE-1. At least one active business is connected. PRE-2. Valid Google tokens exist.
Postconditions	POST-1. New reviews are stored without duplicates. POST-2. UC-07 is triggered for each new review with text.
Normal Flow	6.0 Fetch Reviews 1. Scheduler starts the fetch job. 2. System refreshes Google tokens if expired. 3. System requests new reviews for each active business. (see 6.0.E1) 4. System stores each new review and skips duplicates. 5. System skips reviews that already have a Google reply. 6. System triggers UC-07 for each new review with text.
Alternative Flows	6.1 Manual fetch: the owner triggers the job for one business; continue from step 2.
Exceptions	6.0.E1 Rate limit or Google error 1. System retries with exponential backoff and logs the error. 2. If still failing, system ends the run and tries again at the next schedule.

Business Rules	BR-1 Only active businesses are fetched.
Assumptions	1. Google API quotas are sufficient for the number of businesses.

Table 5.7 Use case description: UC-07 Generate AI Response (Automated)

Use Case ID	UC-07
Use Case Name	Generate AI Response (Automated)
Actors	Primary Actor: Scheduler (Cron Job) Secondary Actors: Google Gemini AI API, Email Service
Description	For each new review, the system classifies sentiment, generates a reply with Gemini, and notifies the owner.
Trigger	UC-06 stores a new review that has text.
Preconditions	PRE-1. A new review with text exists. PRE-2. Response settings exist for the business.
Postconditions	POST-1. Review sentiment and AI reply are stored and status is ai_generated. POST-2. An email notification is sent to the owner.
Normal Flow	7.0 Generate AI Response 1. System classifies the review sentiment. 2. System builds the prompt from business context, settings, and review data. 3. System sends the prompt to the Gemini API. (see 7.0.E1) 4. System stores the reply and sets the status to ai_generated. 5. System sends the owner an email notification. (see 7.0.E2)
Alternative Flows	7.1 Regenerate: the owner requests a new reply from UC-03; the system repeats steps 2 to 4.
Exceptions	7.0.E1 Gemini request fails 1. System retries up to 3 times with exponential backoff. 2. If still failing, the review stays pending for manual reply. 7.0.E2 Email fails 1. System logs the failure; core functionality is unaffected.
Business Rules	BR-1 Replies are 2 to 4 sentences and never offer discounts or free items. BR-2 One notification per review.
Assumptions	1. The Gemini API remains available within free-tier limits.

Table 5.8 Use case description: UC-08 Manage Users

Use Case ID	UC-08
Use Case Name	Manage Users
Actors	Primary Actor: System Administrator Secondary Actors: None
Description	The administrator views user accounts and activates or deactivates them.

Trigger	The administrator opens the user management page.
Preconditions	PRE-1. Administrator is logged in with the admin role.
Postconditions	POST-1. The selected account status is updated.
Normal Flow	8.0 Manage Users 1. Administrator opens the user management page. 2. System displays the list of users. 3. Administrator selects a user and chooses Activate or Deactivate. 4. System updates the account status and confirms.
Alternative Flows	None.
Exceptions	8.0.E1 User not found 1. System displays an error and returns to the list.
Business Rules	BR-1 Only the admin role may access this use case.
Assumptions	1. Administrator accounts are created by the system operator.

4. Specific Requirements

Functional requirements are organized by system feature and written using the functional requirement template.

4.1 System feature 1: Authentication and Account Management

Allows users to create accounts, log in securely, and manage their profile. It is the entry point to every other feature.

Table 6.1 Functional requirement FR-AUTH-01

Identifier	FR-AUTH-01
Title	User Registration
Requirement	The system shall allow a new user to register by providing a name, email address, and password, and shall validate that the email is in a valid format and not already registered.
Source	Business Owner
Rationale	Owners need their own account to keep their businesses and reviews private.
Business Rule (if required)	Email addresses must be unique across all users.
Dependencies	FR-AUTH-02
Priority	High

Table 6.2 Functional requirement FR-AUTH-02

Identifier	FR-AUTH-02
Title	Password Protection
Requirement	The system shall hash every password using bcrypt (minimum 12 salt rounds) before storing it and shall never return the password in any response.
Source	Client
Rationale	Protects credentials if the database is exposed.
Business Rule (if required)	Passwords shall be at least 6 characters long.
Dependencies	FR-AUTH-01
Priority	High

Table 6.3 Functional requirement FR-AUTH-03

Identifier	FR-AUTH-03
Title	User Login

Requirement	When a user submits a valid email and password, the system shall authenticate the user and return a signed JWT; on invalid credentials it shall return a clear error message.
Source	Business Owner
Rationale	Secure access to the dashboard.
Business Rule (if required)	None
Dependencies	FR-AUTH-01
Priority	High

Table 6.4 Functional requirement FR-AUTH-04

Identifier	FR-AUTH-04
Title	Token Validation and Expiry
Requirement	The system shall validate the JWT on all protected endpoints and shall require re-authentication after the token expires (7 days).
Source	System
Rationale	Prevents unauthorized access to owner data.
Business Rule (if required)	A user may only access their own businesses and reviews.
Dependencies	FR-AUTH-03
Priority	High

Table 6.5 Functional requirement FR-AUTH-05

Identifier	FR-AUTH-05
Title	Profile Management
Requirement	The Business Owner shall be able to view and update profile information (name and notification preference).
Source	Business Owner
Rationale	Lets users keep account information current.
Business Rule (if required)	None
Dependencies	FR-AUTH-03
Priority	Medium

4.2 System feature 2: Google Account Linking and Business Management

Connects ReviewPilot to the owner's Google Business Profile and manages the business locations and response settings.

Table 6.6 Functional requirement FR-BIZ-01

Identifier	FR-BIZ-01
Title	Link Google Business Profile
Requirement	When the owner selects Connect Google Business Profile, the system shall redirect the owner to Google's OAuth 2.0 consent screen and, after consent, exchange the authorization code for access and refresh tokens.
Source	Business Owner
Rationale	Grants access to reviews without sharing a Google password.
Business Rule (if required)	Must comply with Google OAuth 2.0 policies and consent screen requirements.
Dependencies	FR-AUTH-03
Priority	High

Table 6.7 Functional requirement FR-BIZ-02

Identifier	FR-BIZ-02
Title	Secure Token Storage and Refresh
Requirement	The system shall store Google tokens encrypted and shall automatically refresh an expired access token using the stored refresh token.
Source	Client
Rationale	Keeps the connection working without repeated logins and protects tokens.
Business Rule (if required)	Tokens shall be encrypted at rest.
Dependencies	FR-BIZ-01
Priority	High

Table 6.8 Functional requirement FR-BIZ-03

Identifier	FR-BIZ-03
Title	Import Business Locations
Requirement	The system shall list the business locations available in the connected profile and allow the owner to select one or more to monitor.
Source	Business Owner
Rationale	Owners may manage several locations.

Business Rule (if required)	None
Dependencies	FR-BIZ-01
Priority	High

Table 6.9 Functional requirement FR-BIZ-04

Identifier	FR-BIZ-04
Title	Display Business Details
Requirement	The system shall display each business's name, address, category, and current average rating.
Source	Business Owner
Rationale	Gives context on the dashboard.
Business Rule (if required)	None
Dependencies	FR-BIZ-03
Priority	Medium

Table 6.10 Functional requirement FR-BIZ-05

Identifier	FR-BIZ-05
Title	Configure Response Settings
Requirement	The Business Owner shall be able to set, per business, the response tone (professional, friendly, casual, or formal), language, sign-off name, and SEO keywords.
Source	Business Owner
Rationale	Makes AI replies match the brand voice.
Business Rule (if required)	Language is limited to English in this version.
Dependencies	FR-BIZ-03
Priority	High

Table 6.11 Functional requirement FR-BIZ-06

Identifier	FR-BIZ-06
Title	Auto-Approve Rule
Requirement	The Business Owner shall be able to enable or disable auto-approve for reviews rated 4 to 5 stars, per business.

Source	Business Owner
Rationale	Reduces effort for routine positive reviews.
Business Rule (if required)	Disabled by default; never applies to reviews below 4 stars.
Dependencies	FR-BIZ-05, FR-RES-05
Priority	Low

Table 6.12 Functional requirement FR-BIZ-07

Identifier	FR-BIZ-07
Title	Deactivate or Remove Business
Requirement	The Business Owner shall be able to deactivate a business (stop fetching reviews) or remove it from the system.
Source	Business Owner
Rationale	Gives control over which locations are monitored.
Business Rule (if required)	None
Dependencies	FR-BIZ-03
Priority	Medium

4.3 System feature 3: Review Fetching

Keeps ReviewPilot synchronized with Google reviews through scheduled and manual fetching.

Table 6.13 Functional requirement FR-REV-01

Identifier	FR-REV-01
Title	Scheduled Review Fetch
Requirement	The system shall fetch new reviews from the Google Business Profile API for every active business every 30 minutes.
Source	Client
Rationale	Ensures timely responses.
Business Rule (if required)	Only active businesses are fetched.
Dependencies	FR-BIZ-02
Priority	High

Table 6.14 Functional requirement FR-REV-02

Identifier	FR-REV-02
Title	Store Reviews Without Duplicates
Requirement	The system shall store the reviewer name, rating, comment, review date, and reviewer photo URL for each review and shall not create a duplicate for an existing Google review ID.
Source	System
Rationale	Prevents repeated replies and notifications.
Business Rule (if required)	Google review ID must be unique.
Dependencies	FR-REV-01
Priority	High

Table 6.15 Functional requirement FR-REV-03

Identifier	FR-REV-03
Title	Skip Already-Answered Reviews
Requirement	The system shall detect reviews that already have a reply on Google and shall not generate a new reply for them.
Source	System
Rationale	Avoids overwriting or duplicating an owner's reply.
Business Rule (if required)	None
Dependencies	FR-REV-02
Priority	Medium

Table 6.16 Functional requirement FR-REV-04

Identifier	FR-REV-04
Title	Manual Fetch
Requirement	The Business Owner shall be able to trigger an immediate review fetch for a selected business.
Source	Business Owner
Rationale	Lets owners check for a new review on demand.
Business Rule (if required)	None
Dependencies	FR-REV-01

Priority	Medium
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Table 6.17 Functional requirement FR-REV-05

Identifier	FR-REV-05
Title	Rate Limit and Error Handling
Requirement	If Google returns a rate-limit or error response, the system shall retry with exponential backoff, log the error, and continue running without crashing.
Source	System
Rationale	Google enforces daily quotas and can be temporarily unavailable.
Business Rule (if required)	None
Dependencies	FR-REV-01
Priority	High

4.4 System feature 4: AI Response Generation

Classifies each review and drafts a personalized reply using the Google Gemini API.

Table 6.18 Functional requirement FR-AI-01

Identifier	FR-AI-01
Title	Sentiment Classification
Requirement	The system shall classify every new review as positive, negative, neutral, or mixed.
Source	Client
Rationale	Determines the style of the reply and feeds analytics.
Business Rule (if required)	None
Dependencies	FR-REV-02
Priority	High

Table 6.19 Functional requirement FR-AI-02

Identifier	FR-AI-02
Title	Generate Reply
Requirement	For each new review with text, the system shall send a prompt containing the business name, category, tone, reviewer name, rating, review text, SEO keywords, and sign-off name to the Google Gemini API and store the returned reply.

Source	Business Owner
Rationale	Core value of the product.
Business Rule (if required)	None
Dependencies	FR-AI-01, FR-BIZ-05
Priority	High

Table 6.20 Functional requirement FR-AI-03

Identifier	FR-AI-03
Title	Reply Style Rules
Requirement	Replies shall be 2 to 4 sentences, sound natural and human, be grateful for positive reviews, apologetic for negative reviews, and balanced for neutral reviews, and shall never offer discounts or free items.
Source	Client
Rationale	Keeps replies safe and appropriate for public posting.
Business Rule (if required)	No discounts or free items may be offered.
Dependencies	FR-AI-02
Priority	High

Table 6.21 Functional requirement FR-AI-04

Identifier	FR-AI-04
Title	Regenerate Reply
Requirement	The Business Owner shall be able to request a new reply for a review, and the system shall replace the draft with the new one.
Source	Business Owner
Rationale	Owners may be unsatisfied with the first draft.
Business Rule (if required)	None
Dependencies	FR-AI-02
Priority	High

Table 6.22 Functional requirement FR-AI-05

Identifier	FR-AI-05
Title	AI Failure Handling

Requirement	If a Gemini request fails, the system shall retry up to 3 times with exponential backoff and, if it still fails, keep the review pending so the owner can reply manually.
Source	System
Rationale	Prevents lost reviews when the AI service is unavailable.
Business Rule (if required)	None
Dependencies	FR-AI-02
Priority	Medium

Table 6.23 Functional requirement FR-AI-06

Identifier	FR-AI-06
Title	Token Usage Tracking
Requirement	The system shall record prompt tokens and completion tokens for each generated reply.
Source	Client
Rationale	Supports cost monitoring against free-tier limits.
Business Rule (if required)	None
Dependencies	FR-AI-02
Priority	Low

4.5 System feature 5: Response Management

Gives the owner control over reviewing, editing, approving, and posting replies.

Table 6.24 Functional requirement FR-RES-01

Identifier	FR-RES-01
Title	List Pending Reviews
Requirement	The system shall display pending reviews together with their AI-drafted replies in a list view.
Source	Business Owner
Rationale	One place to handle all new reviews.
Business Rule (if required)	None
Dependencies	FR-AI-02
Priority	High

Table 6.25 Functional requirement FR-RES-02

Identifier	FR-RES-02
Title	Approve Reply
Requirement	The Business Owner shall be able to approve a drafted reply with a single click.
Source	Business Owner
Rationale	Fast approval workflow (no more than two clicks from viewing a review).
Business Rule (if required)	None
Dependencies	FR-RES-01
Priority	High

Table 6.26 Functional requirement FR-RES-03

Identifier	FR-RES-03
Title	Edit Reply
Requirement	The Business Owner shall be able to edit a drafted reply before approving it.
Source	Business Owner
Rationale	Owners may wish to personalize a reply.
Business Rule (if required)	None
Dependencies	FR-RES-01
Priority	High

Table 6.27 Functional requirement FR-RES-04

Identifier	FR-RES-04
Title	Skip Review
Requirement	The Business Owner shall be able to skip a review so that no reply is posted.
Source	Business Owner
Rationale	Some reviews may not need a reply.
Business Rule (if required)	None
Dependencies	FR-RES-01
Priority	Medium

Table 6.28 Functional requirement FR-RES-05

Identifier	FR-RES-05
Title	Track Response Status
Requirement	The system shall keep each review's status as pending, ai_generated, approved, edited, posted, or skipped and update it at every step.
Source	System
Rationale	Drives the dashboard and prevents duplicate posting.
Business Rule (if required)	A review can be posted only once.
Dependencies	FR-RES-02, FR-RES-03, FR-RES-04
Priority	High

Table 6.29 Functional requirement FR-RES-06

Identifier	FR-RES-06
Title	Post Reply to Google
Requirement	When a reply is approved, the system shall post it to Google through the Business Profile API and mark it as posted only after Google confirms success.
Source	Business Owner
Rationale	Publishes the reply publicly.
Business Rule (if required)	Only approved or edited replies may be posted.
Dependencies	FR-RES-05, FR-BIZ-02
Priority	High

Table 6.30 Functional requirement FR-RES-07

Identifier	FR-RES-07
Title	Response History
Requirement	The system shall display the history of all approved, edited, posted, and skipped responses.
Source	Business Owner
Rationale	Traceability of past replies.
Business Rule (if required)	None
Dependencies	FR-RES-05
Priority	Medium

4.6 System feature 6: Analytics Dashboard

Presents review performance metrics and charts to the owner.

Table 6.31 Functional requirement FR-ANA-01

Identifier	FR-ANA-01
Title	Summary Metrics
Requirement	The system shall display total reviews, pending responses, response rate, and average rating on the dashboard.
Source	Business Owner
Rationale	Quick overview of reputation health.
Business Rule (if required)	None
Dependencies	FR-REV-02, FR-RES-05
Priority	High

Table 6.32 Functional requirement FR-ANA-02

Identifier	FR-ANA-02
Title	Response Rate Calculation
Requirement	The system shall calculate response rate as responded reviews divided by total reviews multiplied by 100.
Source	Client
Rationale	Consistent definition of the key metric.
Business Rule (if required)	Responded means status posted.
Dependencies	FR-ANA-01
Priority	High

Table 6.33 Functional requirement FR-ANA-03

Identifier	FR-ANA-03
Title	Charts
Requirement	The system shall display a line chart of review volume for the last 30 days, a donut chart of sentiment, and a bar chart of rating distribution (1 to 5 stars).
Source	Business Owner
Rationale	Visual insight into trends.

Business Rule (if required)	None
Dependencies	FR-ANA-01, FR-AI-01
Priority	Medium

Table 6.34 Functional requirement FR-ANA-04

Identifier	FR-ANA-04
Title	Filters
Requirement	The Business Owner shall be able to filter analytics by business and by date range, and the data shall refresh when new reviews are fetched or replies are posted.
Source	Business Owner
Rationale	Owners may manage multiple locations.
Business Rule (if required)	None
Dependencies	FR-ANA-03
Priority	Medium

4.7 System feature 7: Email Notifications

Alerts owners when new reviews arrive.

Table 6.35 Functional requirement FR-NOT-01

Identifier	FR-NOT-01
Title	New Review Email
Requirement	When a new review is detected, the system shall send the owner an email containing the reviewer name, star rating, review text, the AI-drafted reply, and a link to the dashboard.
Source	Business Owner
Rationale	Prompt awareness and quick approval.
Business Rule (if required)	None
Dependencies	FR-AI-02
Priority	High

Table 6.36 Functional requirement FR-NOT-02

Identifier	FR-NOT-02
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Title	No Duplicate Notifications
Requirement	The system shall not send more than one notification for the same review.
Source	System
Rationale	Avoids spamming owners.
Business Rule (if required)	One notification per review.
Dependencies	FR-NOT-01
Priority	High

Table 6.37 Functional requirement FR-NOT-03

Identifier	FR-NOT-03
Title	Opt Out
Requirement	The Business Owner shall be able to opt out of email notifications.
Source	Business Owner
Rationale	User control over communication.
Business Rule (if required)	None
Dependencies	FR-AUTH-05
Priority	Low

4.8 System feature 8: Administration

Lets a system administrator manage accounts and monitor the platform.

Table 6.38 Functional requirement FR-ADM-01

Identifier	FR-ADM-01
Title	Manage Users
Requirement	The System Administrator shall be able to view all users and activate or deactivate user accounts.
Source	System Administrator
Rationale	Platform support and abuse control.
Business Rule (if required)	Only users with the admin role may access this feature.
Dependencies	FR-AUTH-04
Priority	Medium

Table 6.39 Functional requirement FR-ADM-02

Identifier	FR-ADM-02
Title	Platform Statistics
Requirement	The System Administrator shall be able to view platform-wide counts of users, businesses, reviews, and generated replies.
Source	System Administrator
Rationale	Monitor system health and usage.
Business Rule (if required)	Only users with the admin role may access this feature.
Dependencies	FR-ADM-01
Priority	Low

5. Quality attributes

5.1 Usability

USE-1: The user interface shall be responsive and usable on screen widths from 375 px to 1920 px.

USE-2: The review approval workflow shall require no more than 2 clicks (view, then approve).

USE-3: The Google connection process shall be completed in fewer than 5 steps.

USE-4: Error messages shall be user-friendly and actionable, not raw error codes.

USE-5: The dashboard shall show skeleton loading states while data is being loaded.

5.2 Performance

PER-1: Standard API operations (excluding AI generation) shall respond within 500 ms for 95% of requests.

PER-2: AI response generation shall complete within 10 seconds per review.

PER-3: The dashboard shall load within 3 seconds on a standard broadband connection.

PER-4: The system shall support at least 100 concurrent users without noticeable degradation.

PER-5: Database queries on reviews shall use indexes on business and date, and on business and status.

5.3 Security

SEC-1: Passwords shall be hashed with bcrypt using at least 12 salt rounds.

SEC-2: All communication shall use HTTPS (TLS 1.2 or later).

SEC-3: JWTs shall be signed with a secret key of at least 256 bits.

SEC-4: Google OAuth tokens shall be stored encrypted in the database.

SEC-5: The system shall validate and sanitize all user input to prevent injection attacks.

SEC-6: CORS shall restrict API access to authorized frontend domains only.

SEC-7: Users shall only access businesses and reviews that belong to their own account.

SEC-8: Secrets shall be kept in environment variables and never committed to version control.

5.4 Reliability

REL-1: The system shall target 99% uptime, excluding scheduled maintenance.

REL-2: The system shall continue to run when the Google API or Gemini API is unavailable.

REL-3: Failed AI generation shall be retried up to 3 times with exponential backoff.

REL-4: The scheduled fetch job shall recover automatically after a failure.

6. External interface requirements

6.1 User interfaces

The user interface is a responsive React web application. Mockups of the major screens are given in Section 16 of the Scope Document. The main screens are listed below.

Table 7 User interface screens

Screen	Description	Key Elements
Login / Registration	User authentication and signup	Email, password, name inputs; submit button
Dashboard	Main overview	Stat cards, review volume chart, pending reviews list
Reviews	List of all reviews	Filters, table with status, rating, date, action
Review Detail	Single review with AI reply	Review text, AI reply, approve/edit/regenerate/skip buttons
Business Settings	Response behavior configuration	Tone selector, sign-off name, keywords, auto-approve toggle
Google Connection	OAuth linking flow	Connect button, status indicator, location selection
Analytics	Detailed charts and metrics	Line, donut, and bar charts; business and date filters
Admin Users	User management (admin only)	User list, activate/deactivate actions

6.2 Software interfaces

Table 8 Software interfaces

External System	Interface Type	Purpose
Google Business Profile API	REST API over HTTPS	Fetch reviews, post replies, get business details
Google OAuth 2.0	OAuth protocol over HTTPS	Authorize access to the Google Business Profile
Google Gemini API	REST API over HTTPS	Classify sentiment and generate replies
MongoDB Atlas	MongoDB wire protocol over TLS	Persistent data storage
SMTP email server	SMTP	Send notification emails

6.3 Hardware interfaces

ReviewPilot has no direct hardware interfaces. Users access the system through a standard computer, tablet, or smartphone with a modern web browser and an internet connection. The server side runs on cloud-hosted infrastructure that is managed by the hosting providers.

6.4 Communications interfaces

All communication between the browser, the backend, and external services shall use HTTPS with TLS 1.2 or later. Notification emails shall be sent using SMTP. Real-time updates over WebSocket may be considered as a future enhancement and are not required in version 1.0.

7. Project Gantt chart

The Gantt chart below shows the planned tasks, durations, and dependencies of the project.

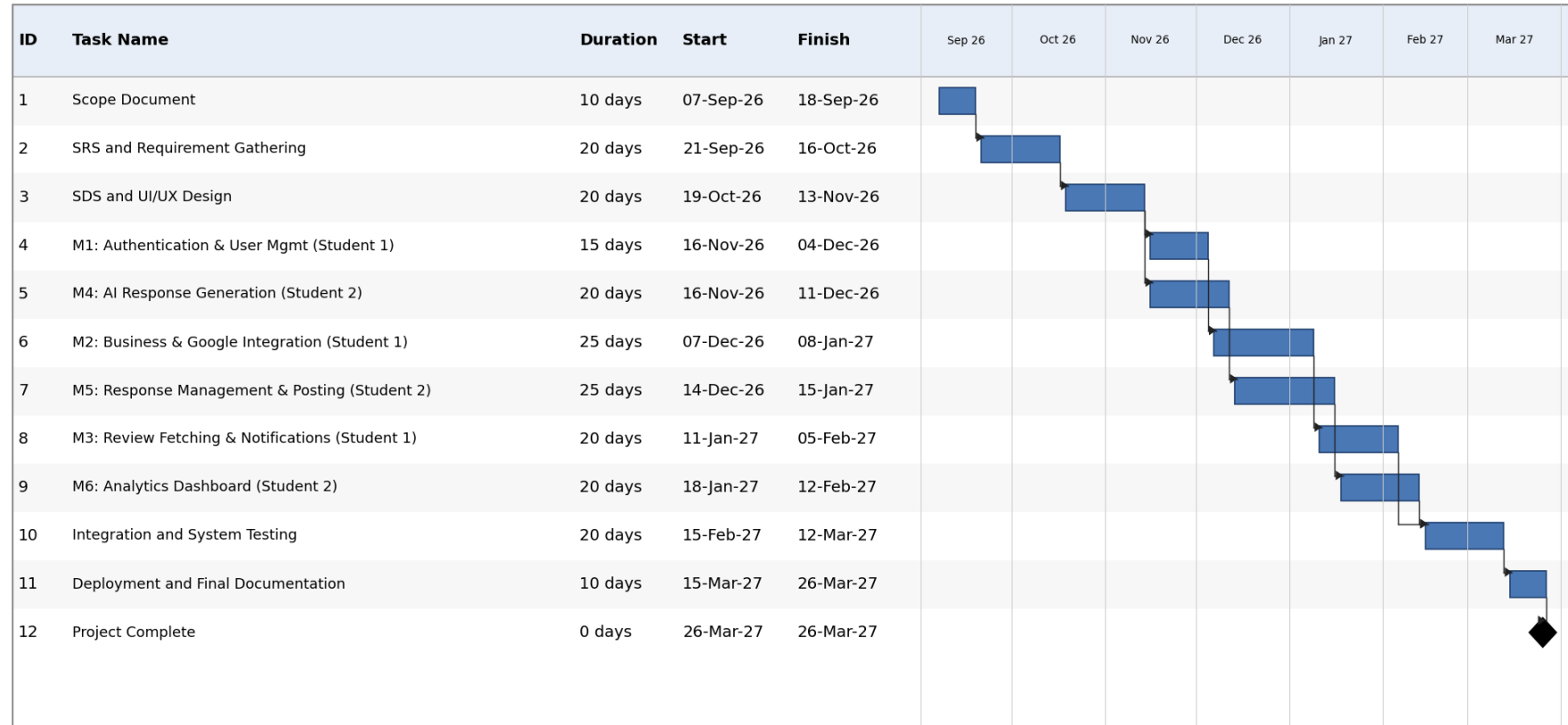


Figure 2 Project Gantt chart

8. References

- [1] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- [2] K. Wiegers and J. Beatty, Software Requirements, 3rd ed. Microsoft Press.
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- [5] OAuth 2.0 Protocol Specification. <https://oauth.net/2/>
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